THE EXTENSION-BY-ZERO CECH COMPLEX IS A RESOLUTION OF THE IMAGE PRESHEAF OF THE AUGMENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author's shorthand (i_0 = i_fix) for the insertion homotopy and the eqnarray* computation are reproduced verbatim.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (the same Stacks-wide preamble of 217 lines that the already published chow and etale-cohomology cards use), namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointer is located at its \label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- The extension-by-zero chain complex K(U)_bullet attached to a covering. Same chapter, Lemma labelled lemma-cech-map-into (source line 1138, Stacks tag 01EL, counted in this batch); named in the counted statement at line 1206. Supplies the complex whose homology presheaves are computed; the computation itself is the counted content.
- Middle of a finite support argument: sections of the extension-by-zero sheaves. Same chapter, the immediately preceding discussion of the presheaves (j_{...})_{p!}O_{...}, together with the statement of the previous lemma. Supplies the simple description of the presheaves (j_{i_0...i_p})_{p!}O_{U_{i_0...i_p}} used at proof lines 1243-1244 to rewrite the terms of the extended complex over an open W.

Source reference. Original lemma 01EL: sources/stacks-cohomology/pinned-original-source.tex, source lines 1138–1138. Same-chapter pointer named in the counted statement at line 1206 for the construction of the complex K(U)_bullet; the section-wise computation of its homology is retained verbatim in the counted span.

Lemma 0.1. *Let X be a ringed space. Let $\mathcal{U} : U = \bigcup_{i \in I} U_i$ be an open covering. Let $\mathcal{O}_{\mathcal{U}} \subset \mathcal{O}_X$ be the image presheaf of the map $\bigoplus j_{p!}\mathcal{O}_{U_i} \to \mathcal{O}_X$. The chain complex $K(\mathcal{U})_\bullet$ of presheaves of Lemma 01EL above has homology presheaves*

$$H_i(K(\mathcal{U})_\bullet) = \begin{cases} 0 & \text{if } i \neq 0 \\ \mathcal{O}_{\mathcal{U}} & \text{if } i = 0 \end{cases}$$

Proof. Consider the extended complex $K^{ext}_\bullet$ one gets by putting $\mathcal{O}_{\mathcal{U}}$ in degree -1 with the obvious map $K(\mathcal{U})_0 = \bigoplus_{i_0} (j_{i_0})_{p!}\mathcal{O}_{U_{i_0}} \to \mathcal{O}_{\mathcal{U}}$. It suffices to show that

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taking sections of this extended complex over any open $W \subset X$ leads to an acyclic complex. In fact, we claim that for every $W \subset X$ the complex $K^{ext}_\bullet(W)$ is homotopy equivalent to the zero complex. Write $I = I_1 \amalg I_2$ where $W \subset U_i$ if and only if $i \in I_1$.

If $I_1 = \emptyset$, then the complex $K^{ext}_\bullet(W) = 0$ so there is nothing to prove.

If $I_1 \neq \emptyset$, then $\mathcal{O}_{\mathcal{U}}(W) = \mathcal{O}_X(W)$ and

$$K^{ext}_p(W) = \bigoplus\nolimits_{i_0 \ldots i_p \in I_1} \mathcal{O}_X(W).$$

This is true because of the simple description of the presheaves $(j_{i_0 \ldots i_p})_{p!}\mathcal{O}_{U_{i_0 \ldots i_p}}$. Moreover, the differential of the complex $K^{ext}_\bullet(W)$ is given by

$$d(s)_{i_0 \ldots i_p} = \sum\nolimits_{j=0,\ldots,p+1} \sum\nolimits_{i \in I_1} (-1)^j s_{i_0 \ldots i_{j-1} i i_j \ldots i_p}.$$

The sum is finite as the element s has finite support. Fix an element $i_{\text{fix}} \in I_1$. Define a map

$$h : K^{ext}_p(W) \longrightarrow K^{ext}_{p+1}(W)$$

by the rule

$$h(s)_{i_0 \ldots i_{p+1}} = \begin{cases} 0 & \text{if} \quad i_0 \neq i_{\text{fix}} \\ s_{i_1 \ldots i_{p+1}} & \text{if} \quad i_0 = i_{\text{fix}} \end{cases}$$

We will use the shorthand $h(s)_{i_0 \ldots i_{p+1}} = (i_0 = i_{\text{fix}}) s_{i_1 \ldots i_p}$ for this. Then we compute

$$\begin{aligned} & (dh + hd)(s)_{i_0 \ldots i_p} \\ = & \sum_j \sum_{i \in I_1} (-1)^j h(s)_{i_0 \ldots i_{j-1} i i_j \ldots i_p} + (i = i_0) d(s)_{i_1 \ldots i_p} \\ = & s_{i_0 \ldots i_p} + \sum_{j \geq 1} \sum_{i \in I_1} (-1)^j (i_0 = i_{\text{fix}}) s_{i_1 \ldots i_{j-1} i i_j \ldots i_p} + (i_0 = i_{\text{fix}}) d(s)_{i_1 \ldots i_p} \end{aligned}$$

which is equal to $s_{i_0 \ldots i_p}$ as desired. $\square$